

Digital Design SET111

Lecture (01)- Course Introduction

Dr. Ahmed Reda Mohamed¹

¹The MoE Key Laboratory of Artificial Intelligence & Bio-Circuits and Systems Laboratory (**BiCASL**), Department of Micro-Nano Electronics, Shanghai Jiao Tong University, Shanghai, China.

- ☐ **Course Information**
- ☐ **Course Assessment**
- ☐ **Course Content**
- ☐ **Course Long-Term-Vision**
- ☐ **IC Industry In Egypt**
- ☐ **References**

- ☐ **Course Information**
- ☐ **Course Assessment**
- ☐ **Course Content**
- ☐ **Course Long-Term-Vision**
- ☐ **IC Industry In Egypt**
- ☐ **References**

Course Information



☐ Lecture's Time:

☐ 10.30: 12.30@ Tuesday

☐ Lecture's Place:

☐ A308

☐ Lectures Materials, Quiz, Assignment, and Project:

☐ Please, check ECU-LMS

☐ Please, check Microsoft team

☐ Contacts:

☐ Email : A.reda@ecu.edu.eg , Ahmed.reda@sjtu.edu.cn

☐ Whatsapp: 01020202427 





- ☐ Course Information
- ☒ Course Assessment
- ☐ Course Content
- ☐ Course Long-Term-Vision
- ☐ IC Industry In Egypt
- ☐ References

Course Assessment



☐ Final Exam [40%]

- ☐ You must get > **30%** of the paper grade or you will **fail** in the course regardless of your course work.

☐ Lab, Quiz, Project, and Assignment [30%]

- ☐ You must attend all course sessions otherwise you will **lose** some degrees

☐ Midterm [30%]

- ☐ 1st midterm [20%] → 7th week
- ☐ 2nd midterm [10%] → 12th week

☐ Project [bonus]

- ☐ 3:5 students



 Forbidden

Copy or cheat during
Exams and even doing the
Quiz and Assignment

- ☐ Course Information
- ☐ Course Assessment
- ☒ Course Content
- ☐ Course Long-Term-Vision
- ☐ IC Industry In Egypt
- ☐ References



□ Course Objective

- Overview of important related topics on digital design systems to expand students' horizons about the digital design of integrated circuit or discrete element.

The course consists of the **six** modules.

□ Module 1: Introductory Concepts.

- Digital and Analog Quantities, Binary Digits, Logic Levels, and Digital Waveforms, Programmable and Fixed-Function Logic Devices.

□ Module 2: Number Systems, Operations, and Codes.

- Decimal, Binary, Octal, Hexadecimal Numbers and their conversion. Complements of Binary Numbers. Logic gates.

□ Module 3: Boolean Algebra and Logic Simplification.

- Laws and Rules of Boolean Algebra, DeMorgan's Theorems, Boolean Expressions and Truth Tables, Karnaugh Map minimization.



☐ Course Objective

- ☐ Overview of important related topics on digital design systems to expand students' horizons about the digital design of integrated circuit or discrete element.

The course consists of the **six** modules.

☐ Module 4: Combinational Logic.

- ☐ Implementing Combinational Logic, Universal NAND and NOR Gates, Adder, Comparators, Decoders, Encoders, Code Converters, Multiplexers and Demultiplexers, Parity Generators/Checkers.

☐ Module 5: Synchronous Sequential Logic

- ☐ Latch, Flip-Flop, Shift Register, Finite State Machine, Counters

☐ Module 6: Programmable Logic

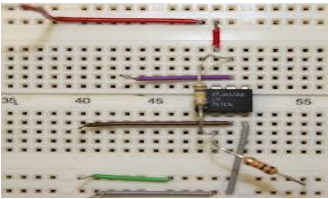
- ☐ Simple Programmable Logic Devices (SPLDs), Complex Programmable Logic Devices (CPLDs), Field-Programmable Gate Arrays (FPGAs)

- ☐ Course Information
- ☐ Course Assessment
- ☐ Course Content
- ☐ **Course Long-Term-Vision**
- ☐ IC Industry In Egypt
- ☐ References

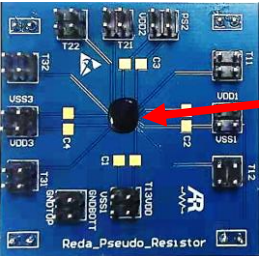


Course Long-Term-Vision (1/3)

Discrete Element



Bread Board

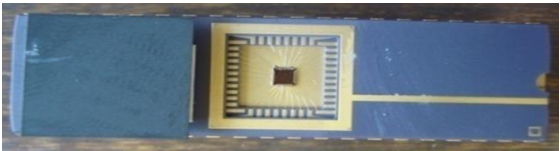


Chip-On-Board (COB)



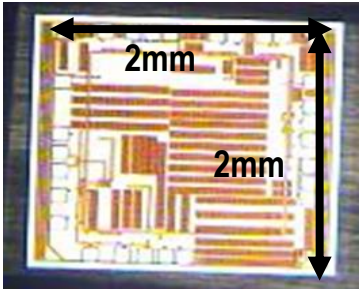
Printed Circuit Board (PCB)

- ☐ Through-hole
- ☐ Surface mount

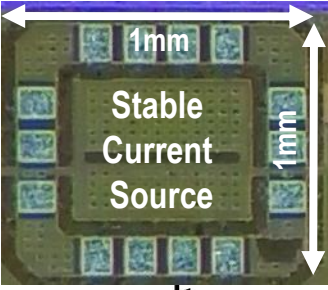


MOSIS (Granted)

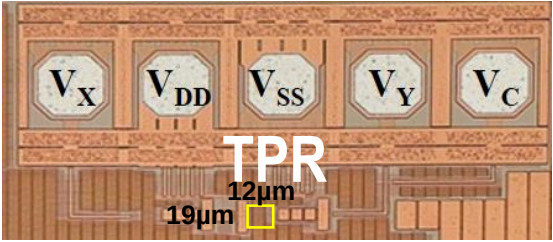
Vey large Scale Integrated Circuit



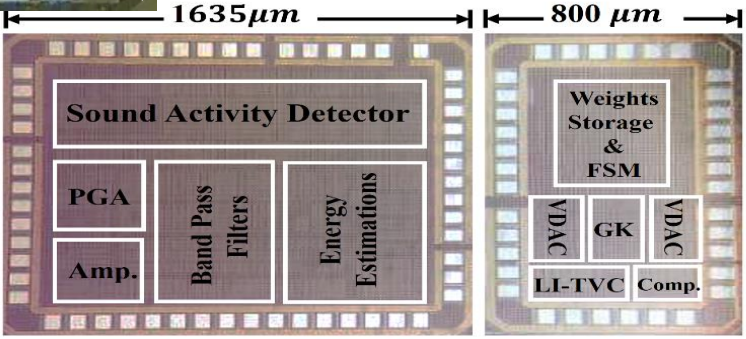
2mm
2mm



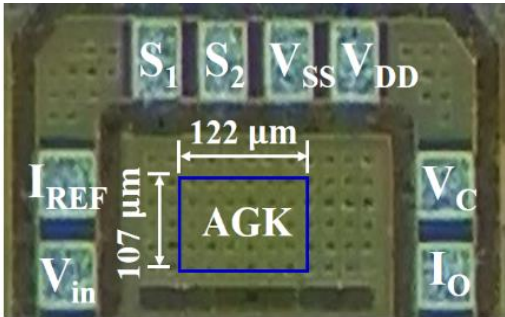
1mm
1mm
Stable Current Source



V_x V_{DD} V_{SS} V_y V_c
TPR
12 μ m
19 μ m



1635 μ m
800 μ m
1420 μ m
Sound Activity Detector
PGA
Band Pass Filters
Energy Estimations
Weights Storage & FSM
V_{DD} GK V_{DD}
V_{DD} LI-TVC Comp.

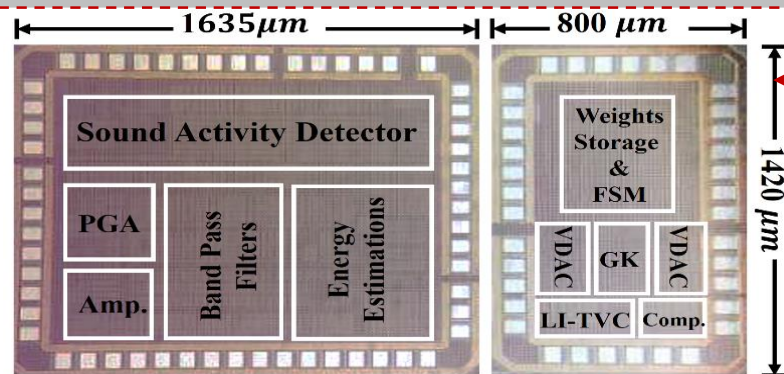
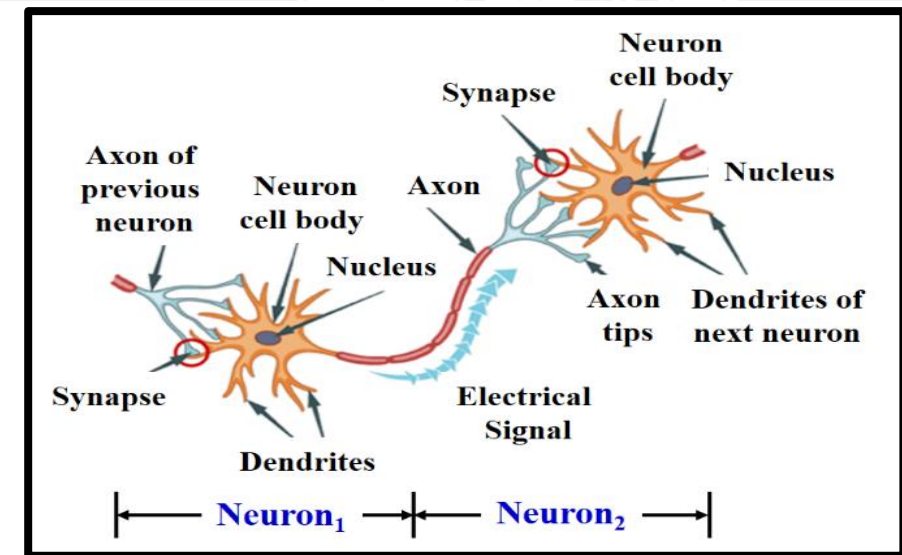
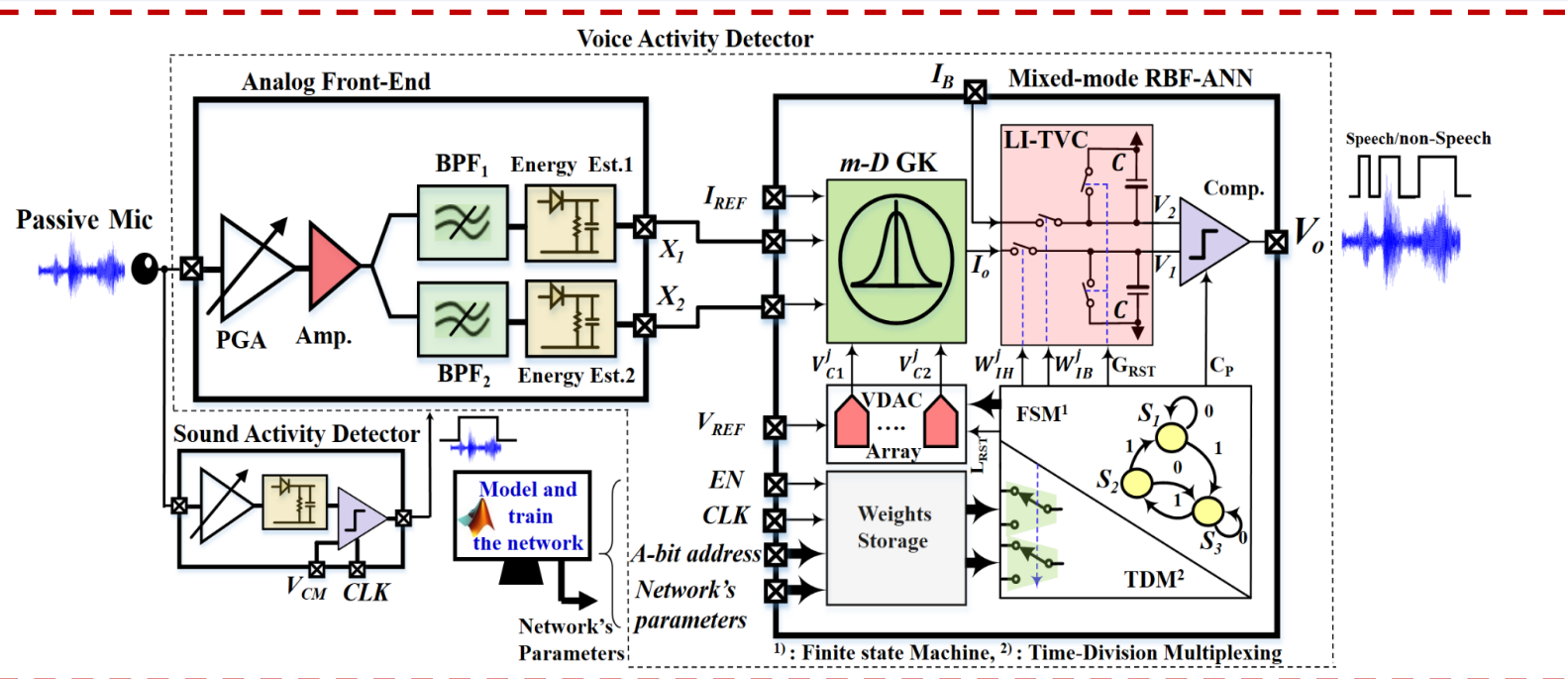


S_1 S_2 V_{SS} V_{DD}
122 μ m
107 μ m
AGK
 I_{REF} V_{in} V_c I_o

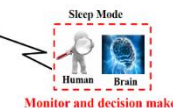
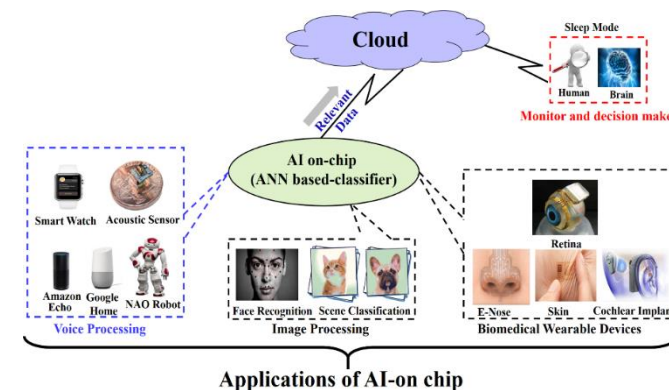
A. Reda

Item	Discrete Element	Very large scale Integrated Circuit
Area	Large	Small
Configuration	Easy	Complex
Price	Cheap	Expensive
Applications	Small production	Mass production
Power consumption	High	Low

Course Long-Term-Vision (2/3)



Artificial Neural Network On-Chip





Electronics in Bio-medical Devices

Pacemaker

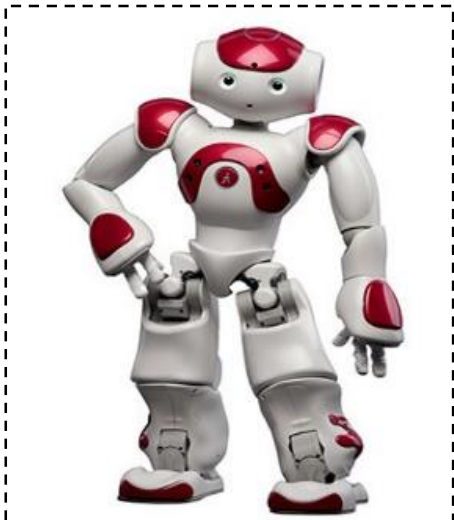
Cochlear implant

ECG monitoring

ECG	ElectroCardio-Gram
PPG	Photo-Plethy-smo Graphy
EEG	Electro-encephalo- gram

PPG monitoring

EEG monitoring



- ☐ Course Information
- ☐ Course Assessment
- ☐ Course Content
- ☐ Course Long-Term-Vision
- ☐ IC Industry In Egypt
- ☐ References

IC Industry In Egypt



- ☐ Course Information
- ☐ Course Assessment
- ☐ Course Content
- ☐ Course Long-Term-Vision
- ☐ IC Industry In Egypt
- ☐ References

References



- ❑ M. M. Mano and M. D. Ciletti, *Digital design With an Introduction to the Verilog HDL*, 5th Ed., Prentice-Hall, Upper Saddle River, NJ, 2013.
- ❑ T. L. Floyd, *Digital Fundamentals*, 11th Ed., Pearson Education Limited, 2015.
- ❑ S. Ramachandran, *Digital VLSI Systems Design A Design Manual for Implementation of Projects on FPGAs and ASICs Using Verilog*, Springer, 2007.

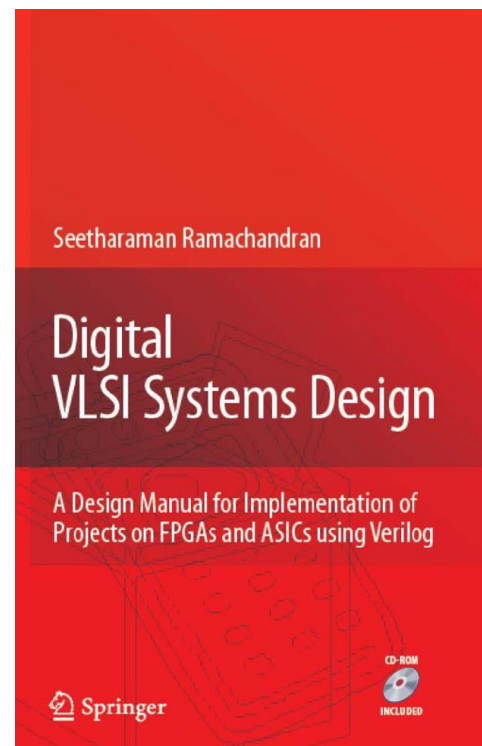
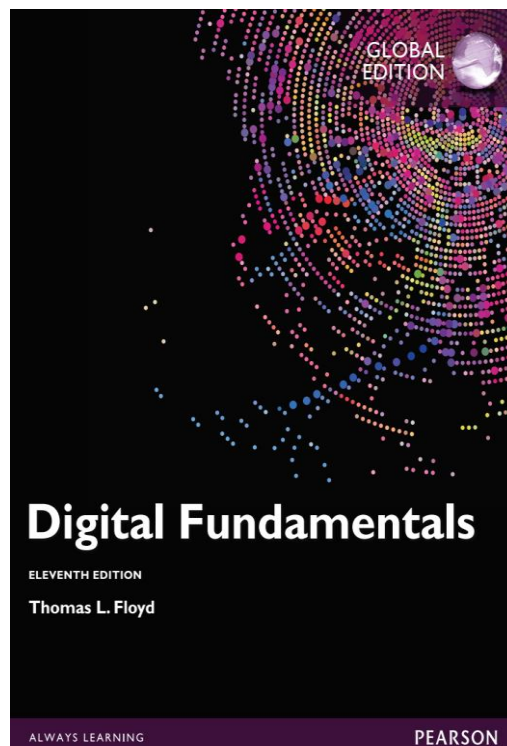
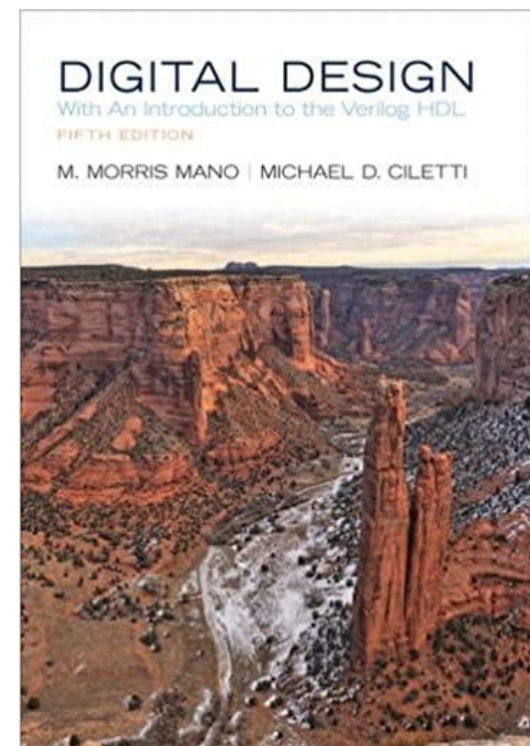
Stay on target
 **[100%]**

❑ **Lectures**

❑ **Sections**

❑ **Labs**

❑ **Project**



Thanks!